LAB 10

Strategy-Search Stage: Grid Search, Out-of-Sample Ranking and Decay

Item	Detail
Topic	Topic 3: Machine Learning Models, Statistical Modelling and A/B Testing for Trading Signals
Pipeline stage	Strategy search
Duration	25 min
Simulator page	http://localhost:8000/#/research/grid?strategy=sma_cross&symbol=SPY
Mode	No coding. Runs ONLINE with your Alpaca paper keys (Integrations page), or OFFLINE on the demo data and the built-in paper broker. Paper trading only.

Lion City Capital and every scenario in these materials are fictitious and for training only. All trading in this course is paper (simulated) trading. Nothing in these materials is investment advice or a recommendation to trade any security.

Scenario

A junior analyst wants the pipeline to 'find the best parameters'. The head of research agrees on one condition: parameters are ranked on 2018-2021 only and judged on the untouched 2022-2025 period (including the 2022 bear market), and the stage must report how much of the in-sample result decays.

Objective

Grid-search SMA crossover parameters in-sample and measure how the ranking decays out of sample.

Before You Start

- Make sure the Trading Simulator is running and open http://localhost:8000 > Course Labs > Lab 10. It is installed once - see the guide 'Download, Install and Start the Trading Simulator Software' in the Learner Guide.
- Record every result you are asked to check - you will use it in the debrief and in the later labs.

Lab Steps - In the Trading Simulator

No coding is needed. Make sure the Trading Simulator is running (installed once - see the guide 'Download, Install and Start the Trading Simulator Software') and open http://localhost:8000. In the sidebar open Course Labs > Lab 10: the app shows the same steps with a Go button that opens each page pre-filled. Every step works offline on the bundled demo data and the built-in paper broker.

Step 1. Run the grid

Research > Strategy search: SMA crossover on SPY, in-sample to 2021-12-31, Alpaca costs. Press Search.

Open: http://localhost:8000/#/research/grid?strategy=sma_cross&symbol=SPY (Research > Grid SPY SMA cross)

Check: About 24 parameter sets; two heatmaps appear.

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Step 2. Read the decay

Compare the in-sample and out-of-sample heatmaps and the IS->OOS rank correlation.

Open: http://localhost:8000/#/research/grid

Check: You can explain why the single best in-sample cell is usually not the best out of sample, and why a robust region is preferred.

Step 3. Walk-forward

Research > Walk-forward: SMA crossover on SPY (3-year train, 1-year test).

Open: http://localhost:8000/#/research/walkforward?strategy=sma_cross (Research > Walk-forward SPY)

Check: The stitched out-of-sample equity curve and the parameters chosen in each window.

Expected in the app: IS and OOS Sharpe heatmaps, a rank correlation and a robust parameter region.

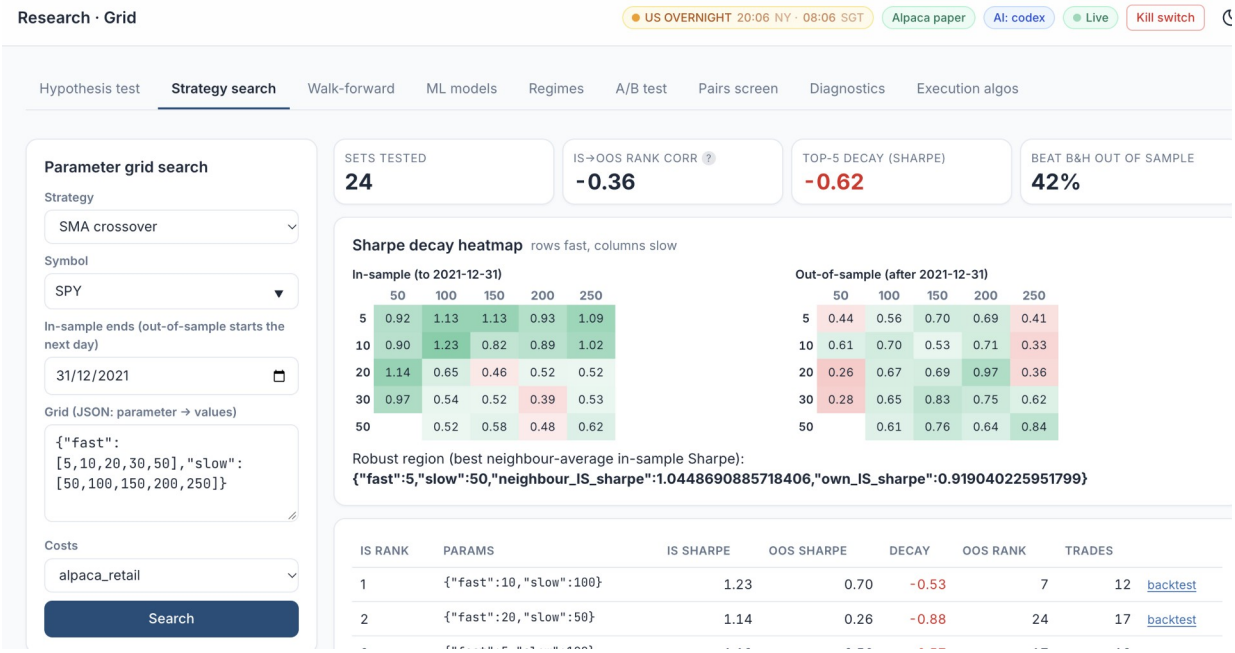


Figure: Lab 10 in the Trading Simulator.

Optional: Let AI Write the Code

This lab needs no coding. To see, run or customise the code behind it, click a link: the AI opens with the prompt filled in and writes the code for you. All prompts are also in labs/Prompt_Library.html and .pdf.

Grid search, out-of-sample ranking and decay - run it in Google Colab
I am a beginner and do not write code. Write ONE complete Python script that runs as-is in a Google Colab notebook. Start with a !pip install line for any library Colab lacks, use pandas, numpy, matplotlib, scipy, scikit-learn, statsmodels and yfinance only, get daily prices from yfinance (auto_adjust=True unless I say otherwise), put a one-line comment above each section, print results as clear tables and charts, and after the code explain in 5 short bullet points how to read the output. Paper trading and education only. Task: grid-search an SMA crossover on SPY (trade at the next day's open with a 0.05% cost per trade): fast 5 to 50 in steps of 5, slow 50 to 250 in steps of 25, fast below slow. Show the Sharpe ratio in-sample (2018-2021) and out-of-sample (2022-2025) as two heatmaps, print the Spearman rank correlation between them, and explain what it says about overfitting.

Open in Claude | Open in ChatGPT | Open Google Colab

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Debrief

Grid search is a statistical search over model structures; comparing in-sample with out-of-sample results exposes how much of the 'best' result was luck. A hold-out period, a count of trials and a preference for robust regions are the pipeline's defences against fooling itself.

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